QUIZ 8-B (EE-1005)

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1. Write the expressions for 3-bits comparator for $\overline{A<B}$, $\overline{A=B}$ and $\overline{A>B}$ [3]

$\overline{A<B}$ = ~~A_3B_3~~ $\bar{A}_2B_2x_3 + \bar{A}_1B_1x_2x_1 + \bar{A}_0B_0x_2x_1x_0$

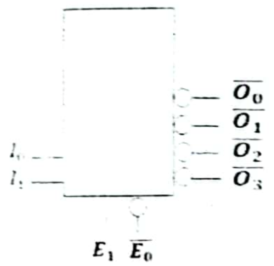
$\overline{A=B}$ = $x_2 \cdot x_1 \cdot x_0$

2

$\because$ $x_2 = A_2 \odot B_2$
$x_1 = A_1 \odot B_1$
$x_0 = A_0 \odot B_0$

$\overline{A>B}$ = $A_2\bar{B}_2x_2 + A_1\bar{B}_1x_2\cancel{x_1} + A_0\bar{B}_0x_2x_1\cancel{x_0}$

2. Produce expressions for **2x4 Decoder** given below? [3]



$\bar{O}_0 = \bar{E}_1 + E_0 + I_1 + I_0$

$\bar{O}_1 = \bar{E}_1 + E_0 + I_1 + \bar{I}_0$

3

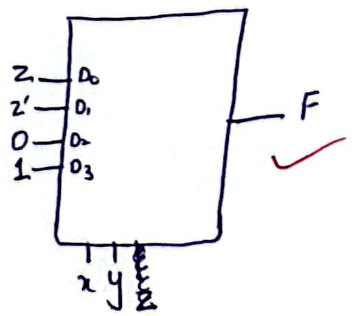
$\bar{O}_2 = \bar{E}_1 + E_0 + \bar{I}_1 + I_0$

$\bar{O}_3 = \bar{E}_1 + E_0 + \bar{I}_1 + \bar{I}_0$

3. Implement given function using **4 X 1 multiplexer?** $F(x,y,z) = \sum(1,2,6,7)$ **Table required** **[5]**

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

5



QUIZ 4-B (EE-1005)

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4. Design Priority **4 x 2** Encoder? Create table, K-map, and expressions? $I_1 > I_3 > I_0 > I_2$ [5]

NOTE: Truth Table must follow **High to Low** priority (High at Left to Lowest at Right)

I_1	I_3	I_0	I_2	O_0	O_1	V
1	x	x	x	0	1	1
0	1	x	x	1	1	1
0	0	1	x	0	0	1
0	0	0	1	1	0	1
0	0	0	0	x	x	0

O_0

I_1I_3 \ I_0I_2	00	01	11	10
00	X	1	0	0
01	1	1	1	1
11	0	0	0	0
10	0	0	0	0

$O_0 = \bar{I}_1 I_3 + \bar{I}_1 I_2 \bar{I}_0$

$O_1 = I_3 + I_1$

5

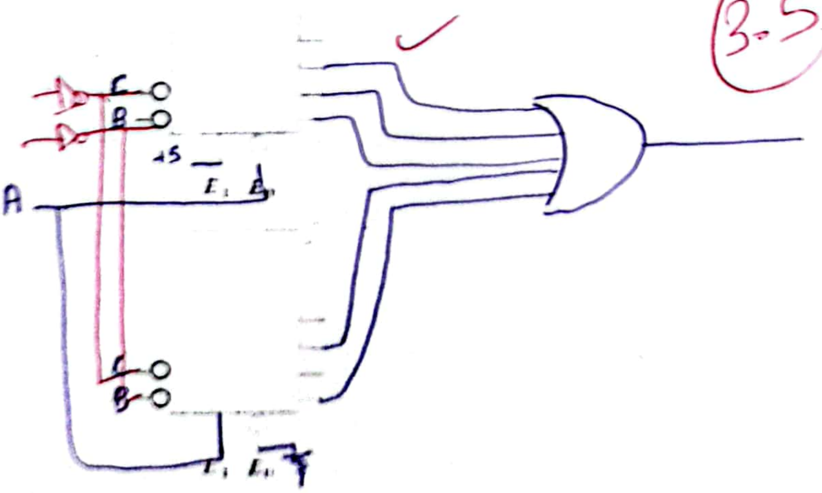
O_1

I_1I_3 \ I_0I_2	00	01	11	10
00	X	0	0	0
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

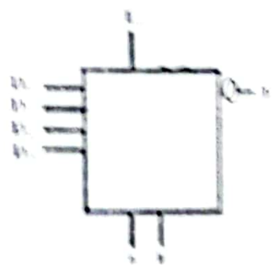
5. Use Decoder to Implement canonical term given below? Rename Inputs according to given function [4]

F(A,B,C)= $\sum(0,1,2,4,6)$

3.5



6. Write the expressions for the **4x1 Multiplexer** given below? [5]



$F = D_0\bar{A}\bar{B}E + D_1\bar{A}BE + D_2A\bar{B}E + D_3ABE$

5

A	B	E	F
0	0	0	x
0	0	1	D_0
0	1	1	D_1
1	0	1	D_2
1	1	1	D_3